Data Structures & Algorithms

An original sample syllabus for exploring Folium. This is illustrative material, not an institutional course document.

01 / Complexity & analysis

Time complexity | Big O notation | Space complexity | Growth rates

A language for comparing algorithms as their inputs grow.

02 / Arrays & linked lists

Contiguous storage | Array operations | Linked nodes | Choosing a structure

Understand how the way we store data shapes the way we use it.

03 / Stacks & queues

LIFO stacks | FIFO queues | Core operations | Practical applications

Control the order in which work is remembered and processed.

04 / Trees & hierarchies

Tree terminology | Binary trees | Search trees | Tree traversals

Represent hierarchical relationships and search through them.

05 / Graphs & networks

Vertices and edges | Graph representations | Breadth-first search | Depth-first search

Model connected systems and explore their relationships.

06 / Searching & sorting

Linear search | Binary search | Merge sort | Algorithm trade-offs

Bring the structures together to find and organize information.